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MILITARY SPECIFICATION

BATTERY RELAY CONTROL UNIT, AIRCRAFT

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for the design, performance, manufacture, and acceptance of a battery relay control unit for use in aircraft to prevent unintentional discharge of aircraft batteries.

1.2 Military part number. The military part number of the battery relay control unit is D85584-1.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 53, Lakehurst, New Jersey 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6140

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2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-15024	Plate, Tags and Bands for Identification of Equipment
DOD-C-38999/42	Connector, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, and Breech Coupling), Environmental Resistant, Removable Crimp and Hermetic Solder Contacts, Receptacle, Box Mounting Flange, Breech Coupling, Removable Crimp Contacts Series IV, Metric
DOD-C-38999/46	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect, (Bayonet, Threaded, and Breech Coupling), Environment Resistant, Removable Crimp and Hermetic Solder Contacts, Plug, Breech Coupling, EMI Grounding, Removable Crimp Contacts, Series IV, Metric
MIL-I-45208	Inspection System Requirements
MIL-I-81023	Inductor, 28 VDC Laboratory Test, General Specification for
MIL-E-81910	Electric Power Generating and Control Equipment, Aircraft, Testing, General Specification for

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electric Component Parts
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference

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MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-2073/1	DOD Packaging Data Forms Instruction for Preparation and Use

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

DEPARTMENT OF DEFENSE

SD-6	Provisions Governing Qualification (Qualified Products List)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Qualification. Battery relay control units furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 Materials.

3.2.1 Threaded parts. All internal or external parts which are threaded shall be positively locked. Accidental loosening of threaded elements shall be prevented by self-locking nuts, safety wiring, or other methods suitable for application in the battery relay control unit.

3.2.2 Fungus. Materials used shall be fungus inert.

3.2.3 Toxic. Materials which produce harmful toxic effects under any conditions, including fire, shall not be incorporated in the design of the battery relay control unit.

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3.2.4 Flammable. Materials which will support combustion shall not be used. Definitions and restrictions on the use of flammable materials shall be in accordance with MIL-STD-454, Requirement 3.

3.2.5 Metals. Metals shall be of a corrosion resisting type or suitably processed to resist corrosion.

3.2.6 Dissimilar metals. The use of dissimilar metals in contact with each other (particularly brass, copper, or steel in contact with aluminum or other alloys) shall be avoided. Where contact between dissimilar metals is unavoidable, they shall be suitably protected against electrolytic corrosion. When protection is used, it shall be a type that offers a low impedance path to radio frequency current. Dissimilar metals shall be as defined in MIL-STD-889.

3.3 Design and construction.

3.3.1 Cooling. The battery relay control unit shall be convection-cooled.

3.3.2 Dimensions. The dimensions shall be in accordance with Figure 1. Where only maximum or minimum dimensions are shown, the part need not have the shape shown, but the part, including all protrusions, shall be contained within the outline shown.

3.3.3 Weight. The maximum weight shall be not greater than 0.3 kilogram.

3.3.4 Mounting. Mounting provisions shall be as shown on Figure 1.

3.3.5 Color. The battery relay control unit shall be finished in 17875 white in accordance with FED-STD-595. The finish shall be unaffected by testing required by this specification.

3.3.6 Electrical connector. The battery relay control unit shall be provided with a D38999/42WB5PN connector in accordance with DOD-C-38999/42 as shown on Figure 1. The connector shall be oriented as shown on Figure 1, with the main key on the same side of the unit as the label. Equipment structure shall not be used as a current path except for radio noise shielding. The connector mounting flange shall be secured to the battery relay control unit by hardware designed to impede removal.

3.3.7 External adjustments. The battery relay control unit shall have no external adjustments.

3.3.8 Input power characteristics. The battery relay control unit shall meet the requirements for utilization equipment as specified in MIL-STD-704, except as specified herein.

3.3.9 Input voltage. The battery relay control unit shall be capable of withstanding voltages in the range of 0 to 33 VDC from pin A to pin B. The load source voltage will be at the same potential as the voltage from pin A to pin B. The unit shall be capable of initiating a ground path from pin C to pin E, with applied voltage in the range of 18 to 33 VDC. It shall be capable of maintaining the ground path with voltages in the range 10 to 33 VDC. There shall be no uncontrolled on-off cycling, oscillation, or other evidence of instability when operated within the specified input voltage range.

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3.3.10 Input current. The maximum DC input current required for operation shall be not greater than .050 ampere. No portion of the load current input at connector pin C shall be used to operate the battery relay control unit.

3.3.11 Duty rating. The battery relay control unit shall be designed for continuous duty.

3.3.12 Maintainability and repairability. The battery relay control unit shall require no maintenance to meet its requirements. The unit shall be constructed to hinder disassembly or access to any internal components. The unit shall not contain any internal provisions, such as connectors, for repair or maintenance. Any visible hardware items shall be of a type designed to impede removal.

3.3.13 Electrical connection sequence. The battery relay control unit shall be designed such that any sequence of application or removal of electrical inputs at the connector pins taken in any order shall not damage the battery relay control unit.

3.3.14 Insulation resistance. When tested as specified in 4.8.8, the battery relay control unit insulation resistance shall be 100 megohms or greater.

3.3.15 Reliability. The battery relay control unit shall have not less than 30,000 Mean Flight Hours Between Failure (MFHBF) while operating under specified conditions for its first 5.5 years of service life including storage.

3.4 Performance.

3.4.1 Electrical performance. The battery relay control unit shall meet the performance requirements specified in 3.4.1.1 through 3.4.1.6 when connected as specified on Figure 2, with input and load source voltage in the range specified by 3.3.9 applied from connector pin A to connector pin B.

3.4.1.1 Ground path. The battery relay control unit connector pin C shall provide a ground path when ground is applied to connector pins D and E. If there is no ground at connector pin E, connector pin C shall be disconnected from ground and there shall be no current greater than 10 microamps through any connector pin. The ground path to be provided by connector pin C shall be a current path to connector pin E; this path shall be capable of conducting 0.75 ampere with a resulting voltage drop not greater than 100 millivolts. The battery relay control unit is not required to initiate the ground path when input operating voltage is below 18.0 VDC.

3.4.1.2 Timing period. If the ground is removed from connector pin D but remains at connector pin E, connector pin C shall provide a ground path in accordance with 3.4.1.1 for 5 $\pm$ 0.5 minutes after which connector pin C shall be disconnected from ground and there shall then be no current greater than 0.005 ampere through any connector pin. If ground is momentarily reapplied to pin D during the 5-minute timing period, the timer shall reset.

3.4.1.3 Reset timing period. If there is no ground at connector pin D and the input voltage at connector pin A is greater than 18.0 VDC and ground is removed from connector pin E for 1.0 millisecond or longer and then reapplied, connector pin C shall provide a ground path in accordance with 3.4.1.1 for

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5.0 ± 0.5 minutes, after which it shall be disconnected from ground and there shall be no current greater than 0.005 ampere through any connector pin.

3.4.1.4 Loss of input voltage. If there is no ground at connector pin D and input voltage is momentarily removed from connector pin A for 60 milliseconds or longer, connector pin C shall be disconnected from ground and there shall be no current greater than .005 ampere through any connector pin.

3.4.1.5 Switching time. The battery relay control unit shall perform its intended switching function within 25 milliseconds of command when input voltage within the range of 3.3.9 is applied. The battery relay control unit shall be capable of switching not less than 0.75 ampere inductive or resistive load. Inductive DC test loads shall use MIL-I-81023 Type II inductors.

3.4.1.6 Ground sensing. When ground is applied to connector pins D and E, there shall be a current of 10 milliamperes or greater through connector pin D. When ground is applied to connector pin E, but no ground is applied to connector pin D, there shall be a voltage of 6 VDC or greater at connector pin D. The values specified shall apply when the input operating voltage at connector pin A is 18.0 VDC or greater.

3.4.1.7 Transient voltage immunity. The battery relay control unit shall be capable of maintaining the ground path from connector pin C to connector pin E while in the timed mode following power interruptions from connector pin A to connector pin B of up to 5.0 milliseconds.

3.4.2 Operating position. The battery relay control unit shall be capable of operating in any orientation or attitude.

3.5 Environmental. The battery relay control unit, when subjected to the environmental conditions listed in Table 1, shall show none of the following.

a. Dimensional distortion beyond specified limits or cracking of cases, covers, or mounting brackets.

b. Breakdown of insulation, loosening or stripping of metal plating or protective coatings from any component part, or corrosion of metal parts.

c. Mechanical failure of any part.

d. Fungus growth.

e. Deterioration of battery relay control unit identification markings.

f. Degradation of electrical performance beyond specified limits of 3.4.

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Table 1. Environmental conditions.

Environmental Condition	Test Paragraph
Acceleration	4.12.1
Dust	4.12.2
Salt Fog	4.12.3
Temperature-Altitude	4.12.4
Mechanical Shock	4.12.5
Fungus	4.12.6
Humidity	4.12.7
Sine Vibration	4.12.8.1
Random Vibration	4.12.8.2
Explosive Atmosphere	4.12.9

3.6 Electromagnetic compatibility. The battery relay control unit shall meet the following electromagnetic compatibility requirements of MIL-STD-461, as specified for category A1b of equipment class A1: CE01, CE03, CE07, CS01, CS02, CS06, RE01, RE02, RS01, RS02, and RS03 except that the RS03 frequencies and field intensities shall be as follows.

Frequency Range	E-Field (Volts/Meter (V/M))
14 KHz - 200 MHz	20
200 MHz - 450 MHz	65
450 MHz - 1.0 GHz	20
1.0 GHz - 12.0 GHz	100
12.0 GHz - 40 GHz*	20

*Except 100 V/M at 16.5 GHz and 33.0 GHz.

3.7 Marking. Markings shall be in accordance with MIL-STD-130.

3.8 Nameplate. Each battery relay control unit shall contain one or more appropriately sized type A, B, or H nameplates, in accordance with MIL-P-15024, securely attached to the unit. Alternate nameplates are acceptable so long as the legibility and durability requirements of this specification are complied with. The manufacturer's name or trademark shall be in letters not larger than other letters appearing on the nameplates. The nameplates shall contain the following information.

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Aircraft Battery Relay Control Unit
Input: 28 volts DC .050 ampere DC (Max)
D85584-1
5945-01-207-9035
Manufacturer's part number
Contract or order number
Relay Control Unit S/N
Date manufactured
Manufacturer's name or trademark
Inspector's stamp
US
Warranty expiration date
THIS UNIT IS NON-REPAIRABLE. DO NOT ATTEMPT TO DISASSEMBLE

3.9 Workmanship. Battery relay control units shall be manufactured and assembled in accordance with MIL-STD-454, requirement 9.

3.10 Soldering. Soldering shall comply with MIL-STD-454, requirement 5.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows.

a. Qualification inspection (see 4.4).

b. Qualification verification inspections (see 4.5).

c. Quality conformance inspection (see 4.7).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.3.1 Inspection system requirements. The inspection system requirements shall be in accordance with MIL-I-45208.

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4.4 Qualification inspection. Qualification inspection will be performed by the Government in accordance with SD-6, Provisions Governing Qualification (see 6.3 and 6.3.1) on samples of battery relay control units furnished by the supplier. Qualification tests shall consist of the examinations, inspections, and tests specified in Table 2 and shall be performed in the order shown.

4.4.1 Qualification samples. The qualification test samples shall be produced, to the maximum extent practicable, with equipment and procedures normally used in production, and shall be identified as specified (see 3.7 and 3.8). Any additional information required by the letter of authorization shall be supplied (see 6.3 and 6.3.1).

4.4.2 Materials certification. Together with the test samples, the manufacturer shall prepare a materials inspection report consisting of certification, supported by verifying data, that the materials and finishes specified in 3.2 used in fabricating the battery relay control units are in accordance with the applicable referenced specifications or specified requirements.

4.5 Verification of qualification. Each battery relay control unit of the same part number delivered as a qualified item under this specification shall be numbered sequentially in the order that it is delivered. To meet the delivery requirement for the first 50 battery relay control units, 51 shall successfully complete the required quality conformance inspections, 50 of which shall be shipped as directed by the applicable contract(s), and 1 battery relay control unit, selected at random by the Government inspector, shall be forwarded to the qualifying activity (see 6.3 and 6.3.1). To meet the delivery requirement for each succeeding 150 battery relay control units, 151 battery relay control units shall successfully complete the required quality conformance inspections, 150 of which shall be shipped as directed by the applicable contract(s), and 1 battery relay control unit selected at random shall be forwarded as above. The span of serial numbers and the contract numbers of the production units from which each verification sample was selected shall be recorded and this information included with the sample. The sample shall be shipped no later than the final unit in the production group from which selected. The units forwarded to the qualifying activity will be inspected and tested for conformance to this specification. Tests will be conducted as specified in Table 2 for any one of the qualification verification samples at the discretion of the qualifying activity. All qualification verification samples become Government property when accepted by the Government inspector for shipment to the qualifying activity.

4.5.1 Rejection and retest of qualification and qualification verification samples. Samples which have been rejected or returned to the manufacturer for any reason during qualification or qualification verification tests may be reworked or have parts replaced to correct defects. Before resubmitting the samples, full particulars concerning the rejection and the corrective action taken by the manufacturer shall be submitted in writing by the manufacturer to the qualifying activity. Tests shall not be resumed until such a report is received. If the manufacturer does not correct and return components rejected during qualification verification testing within 5 weeks after he receives them to the qualifying activity or does not submit a program acceptable to the qualifying activity for the correction of deficiencies noted within 2 weeks, the component will be disqualified at the discretion of the qualifying activity.

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4.5.2 Retention of qualification. In order to retain qualification of a product approved for listing on the Qualified Products List (QPL), the manufacturer shall verify by certification to the qualifying activity that the manufacturer's product complies with the requirements of this specification. The time of periodic verification by certification shall be in 2-year intervals from the date of original qualification. The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specification requirements.

4.6 Inspection. At the conclusion of tests, each sample shall be inspected for evidence of excessive deterioration or defects. The presence of such evidence shall be cause for rejection.

Table 2. Schedule of qualification tests.

Test Title	Battery Relay Control Unit Sample No 1	2	3	Requirement Paragraph	Test Paragraph
Input Power Characteristics		x		3.3.8	4.9.2
Input Voltage	x	x	x	3.3.9	4.9.3
Input Current	x	x	x	3.3.10	4.9.4
Electrical Connection Sequence	x	x	x	3.3.13	4.9.5
Insulation Resistance	x	x	x	3.3.14	4.8.8
Electrical Performance	x	x	x	3.4.1	4.10
Ground Path	x	x	x	3.4.1.1	4.10.1
Timing Period	x	x	x	3.4.1.2	4.10.2
Reset Timing Period	x	x	x	3.4.1.3	4.10.3
Loss of Input Voltage	x	x	x	3.4.1.4	4.10.4
Switching Time		x		3.4.1.5	4.10.5
Ground Sensing			x	3.4.1.6	4.10.6
Transient Voltage Immunity			x	3.4.1.7	4.10.7
Operating Position	x			3.4.2	4.11
Acceleration	x			3.5	4.12.1
Dust	x			3.5	4.12.2
Salt Fog	x			3.5	4.12.3
Temperature-Altitude		x		3.5	4.12.4
Mechanical Shock		x		3.5	4.12.5
Fungus		x		3.5	4.12.6
Humidity			x	3.5	4.12.7
Sine Vibration	x			3.5	4.12.8.1
Random Vibration			x	3.5	4.12.8.2
Explosive Atmosphere			x	3.5	4.12.9
Electromagnetic Compatibility			x	3.6	4.13
Marking	x	x	x	3.7	4.9.1
Nameplate	x	x	x	3.8	4.9.1
Visual and Mechanical Inspection	x	x	x	3.2.3.3	4.9.1

4.7 Quality conformance inspections. Quality conformance inspections shall consist of inspection of the product for delivery. Each battery relay control unit offered for delivery shall be subjected to the inspection

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specified in Table 3. Data from these inspections shall be recorded and available for examination by authorized Government representatives.

4.7.1 Noncompliance. If a sample fails to pass quality conformance inspection, the manufacturer shall notify the procuring activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the procuring activity, has been taken. After the corrective action has been taken, quality conformance inspection shall be repeated on additional sample units (all tests and examinations or the test which the original sample failed, at the option of the procuring activity). Final acceptance and shipment shall be withheld until the quality conformance inspection has shown that the corrective action was successful. In the event of failure after reinspection, further rework and resubmission shall not be allowed.

Table 3. Quality conformance inspection.

Test Title	Requirement Paragraph	Test Paragraph
Insulation Resistance	3.3.14	4.8.8
Electrical Performance	3.4.1	4.10 thru 4.10.5
Visual and Mechanical Inspection	3.2 and 3.3	4.9.1
Marking	3.7	4.9.1
Nameplate	3.8	4.9.1

4.8 Methods of examination and test.

4.8.1 Test conditions. Unless otherwise specified herein, tests shall be made under the conditions specified in 4.8.2 through 4.8.3.

4.8.2 Ambient air. Tests shall be performed at an ambient temperature of 23° ± 5°C at ambient atmospheric pressure and relative humidity.

4.8.3 Tolerance and accuracy. The tolerance of test conditions other than room ambient conditions and the accuracy of test apparatus shall be as specified in MIL-E-81910. Measurements shall be in accordance with MIL-E-81910.

4.8.4 Ambient temperature measurement. When tests are required to be conducted at reduced or elevated temperatures, the ambient temperature shall be determined by measurements of the air at several points surrounding, but not above the unit, and averaging the results.

4.8.5 Airflow. For all high temperature-altitude tests below 6.1 kilometers, the airflow over the unit within the test chamber shall be adjusted to approximate natural convection conditions and shall be not greater

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than 30.5 meters/minute while maintaining the required ambient temperature within specified tolerance. For tests above 6.1 kilometers altitude, the airflow may exceed 30.5 meters/minute but shall be adjusted to the minimum velocity consistent with maintaining the required ambient temperature within specified tolerance.

4.8.6 Position. Unless otherwise specified herein, the units shall be mounted base down.

4.8.7 Input voltage. Nominal or other specified input voltage shall be applied as indicated herein. Nominal voltage shall be 28 volts DC.

4.8.8 Insulation resistance. The battery relay control unit insulation resistance shall be measured between all connector pins and the unpainted electrical grounding area at the case mounting holes in accordance with MIL-STD-202, Method 302, Test Condition B. The ambient relative humidity shall be 30 to 50 percent.

4.9 Methods of inspection.

4.9.1 Visual and mechanical inspection. The battery relay control unit shall be inspected to determine compliance with the requirements of this specification with respect to materials, parts, workmanship, dimensions, weight, markings, and connector orientation.

4.9.2 Input power characteristics. The battery relay control unit shall be tested for conformance with the requirements of 3.3.8. The following transients shall be applied to the battery relay control unit connector pin A while the unit is operating in accordance with 3.4.1.1 and 3.4.1.3.

a. 50 volts for 12.5 milliseconds.

b. 46 volts for 25 milliseconds.

c. 40 volts for 47.5 milliseconds.

The unit shall not be damaged during this test.

4.9.3 Input voltage.

4.9.3.1 Voltage limits for initiating the ground path. The battery relay control unit shall be tested to verify capability of initiating the ground path in accordance with 3.4.1.1 for input voltages of 33.0 VDC, nominal, and 18.0 VDC. The battery relay control unit is not required to initiate the ground path when input voltage is below 18.0 VDC.

4.9.3.2 Voltage limits for maintaining the ground path. The battery relay control unit shall be tested to verify capability of maintaining the ground path in accordance with 3.4.1.1 for input voltages of 33.0 VDC, nominal, and 10.0 VDC.

4.9.3.3 Voltage limits for terminating the ground path. The battery relay control unit shall be tested to verify capability of terminating the ground path in accordance with 3.4.1.1 for input voltages of 33.0 VDC, nominal, and 10.0 VDC.

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4.9.4 Input current. Input voltage shall be applied to connector pin A. No voltage shall be applied to the load at connector pin C. Ground shall be applied continuously to connector pin B and as necessary to connector pins D and E. The input operating current at connector pin A shall be monitored for all functional operations of 3.4.1.1, 3.4.1.2, and 3.4.1.3 to verify that it does not exceed 0.050 ampere.

4.9.5 Electrical connection sequence. Connect the battery relay control unit to input voltage through connector pin A, to ground through connector pin B, to a 0.75 ampere inductive DC load source through connector pin C, and to ground as necessary through connector pins D and E. Inductive DC test loads shall use MIL-I-81023 inductors. Operate the battery relay control unit in each of the modes of 3.4.1 and subparagraphs. Remove and then reapply the electrical inputs at each connector pin taken in random sequence during operation in each of the modes of 3.4.1 and subparagraphs. The battery relay control unit shall meet the requirements of 3.3.13, 3.4.1, and subparagraphs.

4.10 Electrical performance. With the battery relay control unit connected as shown on Figure 1, electrical operation shall be evaluated as specified in 4.10.1 through 4.10.7. Unless otherwise specified herein, input voltage shall be applied from connector pin A to connector pin B, and a 0.75 ampere inductive load conforming to MIL-B-81023, Type II, shall be applied to connector pin C. An appropriate mating plug connector conforming to DOD-C-38999/46 (such as D38999/46WB5SN) shall be used. The battery relay control unit shall be tested over the full input voltage range and ambient temperature range.

4.10.1 Ground path. Apply ground to connector pins D and E. Verify that connector pin C provides a path to ground in compliance with 3.4.1.1. Remove ground from connector pin E. Verify that connector pin C is disconnected from ground and there is no current greater than 10 microamps through any connector pin.

4.10.2 Timing period. Apply ground to connector pins D and E. Remove the ground at connector pin D and begin timing. Verify that connector pin C provides a ground path per 3.4.1.1 for 5 $\pm$ 0.5 minutes. Verify that after 5 $\pm$ 0.5 minutes, connector pin C is disconnected from ground and there is no current greater than 0.005 ampere through any connector pin. Reapply ground to connector pin D. Verify that connector pin C provides a ground path in accordance with 3.4.1.1.

4.10.3 Reset timing period. Apply ground to connector pins D and E. Remove the ground from connector pin D. While connector pin C is providing a ground path, momentarily remove ground from connector pin E for a minimum of 1.0 millisecond. Verify that connector pin C provides a ground path in accordance with 3.4.1.1 for 5 $\pm$ 0.5 minutes from the time of the power interruption; and, that after the timing period, there is no current greater than 0.005 ampere through any connector pin.

4.10.4 Loss of input power. Apply ground to connector pin E. Verify that a ground path is provided at connector pin C. Momentarily remove power from connector pin A for 60 milliseconds minimum. Verify that connector pin C is disconnected from ground and there is no current greater than 0.005 ampere through any connector pin. Apply ground to connector pin D. Verify that a ground path is provided by connector pin C in accordance with 3.4.1.1. Remove power from connector pin A. Verify that connector pin C is disconnected from

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ground and there is no current greater than 10 microamps through any connector pin.

4.10.5 Switching time. With connector pin D open, apply ground to connector pin E. Verify that a ground path is provided by connector pin C within 25 milliseconds.

4.10.6 Ground sensing. Apply ground to connector pins D and E. With an input of 18 volts applied from connector pin A to pin B, verify that the current through connector pin D is not less than 10 milliamps. Remove ground from connector pin D. Verify that the voltage at connector pin D is not less than 6.0 volts, as required by 3.4.1.6.

4.10.7 Transient voltage immunity. With input voltage in the range of 18 to 33 volts applied from pin A to pin B, and pins D and E initially open, apply ground to pin E, establishing a ground path at connector pin C. Momentarily remove input voltage for up to 5.0 milliseconds. The ground path shall be maintained from pin C to pin E following the power interruption.

4.11 Operating position. With the battery relay control unit operating under nominal input conditions, place the unit on each of its six surfaces for 30 minutes each. Confirm that operation of the battery relay control unit in any position does not affect current input, heating, switching, timing, or other operating characteristics.

4.12 Environmental.

4.12.1 Acceleration. The battery relay control unit shall be tested in accordance with the acceleration test of MIL-E-81910, except that the acceleration level shall be 15g. During Procedure II, the battery relay control unit shall operate at its rated load with nominal input voltage applied. The current through each connector pin shall be monitored during the test and shall be free of abnormal interruptions and transients. At the conclusion of the test, the battery relay control unit shall be examined for compliance with 3.5 and shall meet the requirements of 3.4.1 and subparagraphs.

4.12.2 Dust. The battery relay control unit shall be tested in accordance with the dust test of MIL-E-81910, except that no input voltage shall be applied during the test. At the conclusion of the test, the battery relay control unit shall meet the requirements of 3.4.1 and subparagraphs.

4.12.3 Salt fog. The battery relay control unit shall be tested in accordance with the salt fog test of MIL-E-81910. The unit shall then stand at room ambient conditions for a period of 24 hours and shall meet the requirements of 3.4.1 and subparagraphs. After an additional 48-hour stand at room ambient conditions, the unit shall be examined for compliance with 3.5 and shall meet the requirements of 3.4.1 and subparagraphs.

4.12.4 Temperature-altitude. The battery relay control unit shall be tested in accordance with the temperature-altitude test of MIL-E-81910 for self-cooled equipment, except that the maximum and minimum test temperatures shall be 71°C and -55°C, respectively. The maximum test altitude shall be 20 kilometers. The unit shall be electrically evaluated at 4-kilometer intervals. The unit shall meet the requirements of 3.4.1 and subparagraphs.

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4.12.5 Mechanical shock. The battery relay control unit shall be tested in accordance with the mechanical shock test of MIL-E-81910, except that the equipment crash hazard test, Procedure V of MIL-STD-810, shall not be performed. At the conclusion of the test, the unit shall meet the requirements of 3.4.1 and subparagraphs.

4.12.6 Fungus. The battery relay control unit shall be tested in accordance with the fungus test of MIL-E-81910. At the conclusion of the test, the battery relay control unit shall be inspected for fungus growth and shall meet the requirements of 3.4.1 and subparagraphs.

4.12.7 Humidity. The battery relay control unit shall be tested in accordance with the humidity test of MIL-E-81910. The unit shall complete the electrical performance tests while the unit is at 30°C during each cycle of the humidity test. The battery relay control unit shall comply with 3.4.1 and subparagraphs. At the conclusion of the test, the battery relay control unit shall be examined for compliance with 3.5 and shall meet the requirements of 3.4.1 and subparagraphs.

4.12.8 Vibration.

4.12.8.1 Sine vibration. The battery relay control units shall be tested in accordance with the sine vibration test of MIL-E-81910 for non-engine-mounted components, except that the sine vibration test displacements, acceleration levels, and crossover frequencies shall be as follows.

a.	5-14 Hz	5.08 mm Double Amplitude (DA)
b.	14-33 Hz	2 g
c.	33-74 Hz	0.9144 mm DA
d.	74-2000 Hz	10 g

During each axis of vibration, the unit shall comply with 3.4.1 and the unit output waveform shall be free of abnormal interruptions and transients. At the conclusion of the test, the unit shall be examined for compliance with 3.5 and shall meet the requirements of 3.4.1 and subparagraphs.

4.12.8.2 Random vibration. The battery relay control unit shall be tested in accordance with the random vibration test of MIL-E-81910 for non-engine-mounted components, except that the unit shall only be subjected to the functional level of random vibration and the test time duration shall be 2 hours for each axis. During each axis of vibration, the battery relay control unit shall comply with 3.4.1 and the unit output waveform shall be free of abnormal interruptions and transients. At the conclusion of the test, the unit shall be examined for compliance with 3.5 and shall meet the requirements of 3.4.1 and subparagraphs.

4.12.9 Explosive atmosphere. The battery relay control unit shall be subjected to the explosive atmosphere test of MIL-STD-810, Method 511, Procedure I. The maximum chamber temperature shall be not greater than 71°C and the maximum test altitude shall be 20 kilometers. During the test, the battery relay control unit shall be operated to actuate all making and breaking of electrical contacts.

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4.13 Electromagnetic compatibility. The battery relay control unit shall be subjected to the applicable test methods and procedures of MIL-STD-462.

5. PACKAGING.

5.1 Preservation packaging. Preservation packaging shall be in accordance with MIL-STD-2073/1 level A or C, as specified in the contract or purchase order (see 6.2).

5.1.1 Level A. For level A packaging, preservation shall be in accordance with MIL-P-116, Method II, without preservation compounds, using metal reusable containers unless otherwise specified in the contract or purchase order.

5.1.2 Level C. For level C, the battery relay control unit shall be preserved and packaged individually in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be in accordance with MIL-STD-2073/1, level A, B, or C as specified in the contract or purchase order.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-2073/1 and MIL-STD-129.

6. NOTES.

6.1 Intended use. The battery relay control unit is intended for use in aircraft for the purpose of preventing inadvertent discharge of the aircraft storage battery when the aircraft is shut down. To accomplish this purpose, the unit is to automatically cause the aircraft battery relay to electrically disconnect the aircraft battery from the electrical bus after sufficient time has elapsed for normal use of the battery when the aircraft is shut down on the ground.

6.2 Ordering data. Acquisition documents should specify the following.

a. Title, number, and date of this specification.

b. Military part number (see 1.2).

c. Required level of packaging and packing (see 5.1).

d. Type of container (see 5.1.1).

e. Qualification verification samples (see 4.5).

6.3 Qualification. Qualification will be in accordance with SD-6, Provisions Governing Qualification. Awards will be made only for products which are qualified at the time set for opening of bids. The activity responsible for the Qualified Products List is the Naval Air Systems Command (Attention: Commanding Officer, Naval Air Engineering Center (SESD) (Code 53), Lakehurst, NJ 08733-5100).

6.3.1 Application for qualification. The Naval Weapons Support Center, Crane, Indiana, has been designated by the Naval Air Systems Command as the qualifying activity for the establishment of the QPL. Requests for information pertaining to and applications for qualification should be addressed to:

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Commanding Officer, Naval Weapons Support Center (Code 3052), Crane, Indiana 47522-5000.

6.4 Engineering changes to a qualified item. Once a manufacturer's part number has been placed on the Qualified Products List, the manufacturer is responsible for any changes to the battery relay control unit which affect its physical or performance characteristics. No changes should be made to a qualified unit without the specific written approval of the qualifying agency. The manufacturer should allow sufficient lead time for the qualifying agency to repeat the qualification tests if necessary.

6.5 Installation. The battery relay control unit will be installed in the aircraft in accordance with MIL-E-7080 such that the length of unprotected cable between the battery relay control unit and the aircraft battery relay is less than 1 foot.

6.5.1 Installation clearances. Clearance should be provided for installation of battery relay control unit mounting hardware and for connection with mating plug connector D38999/46WB5SN.

6.6 Design guides. The following design guidelines based on actual experience should be addressed and considered by the contractor when a battery relay control unit is to be developed and proposed in accordance with this specification.

6.6.1 Environmental stress screening. Environmental stress screening of electronic parts should be used where required to achieve the MFHBF as specified in 3.3.15.

6.6.2 Semiconductor and electron tubes. Semiconductors should be chosen and applied as outlined in MIL-STD-454, requirement 30. Electron tubes of either the vacuum or gas filled types should not be used.

6.6.3 Internal electric connections. Internal connections should be made in accordance with MIL-STD-454, requirements 1 and 69.

6.6.4 Adjustable resistors. Only sealed adjustable resistors conforming to MIL-R-39015 should be used.

6.6.5 Microelectronic devices. Microelectronic devices should be chosen and applied in accordance with MIL-STD-454, requirement 64 and should be listed in MIL-STD-1562.

6.6.6 Capacitors. Capacitors should be chosen and applied in accordance with MIL-STD-454, requirement 2. The use of wet slug tantalum capacitors is not recommended.

6.7 Manufacturing guides. The appropriate technical requirements contained in the following list of documents should be addressed and considered by the manufacturer where applicable.

Document	Title
MIL-STD-275	Printed Wiring for Electrical Equipment
QQ-P-416	Plating, Cadmium (Electrodeposited)

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MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-C-5541	Chemical Conversion Coating on Aluminum and Aluminum Alloys
MIL-C-6021	Castings, Classification and Inspection of
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-A-21180	Aluminum Alloy Castings, High Strength
MIL-P-55110	Printed Wiring Boards
WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections
NAV MAT-P-9492	Navy Manufacturing Screening Program

6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Subject term (key word) listing.

Battery Relay Control Unit

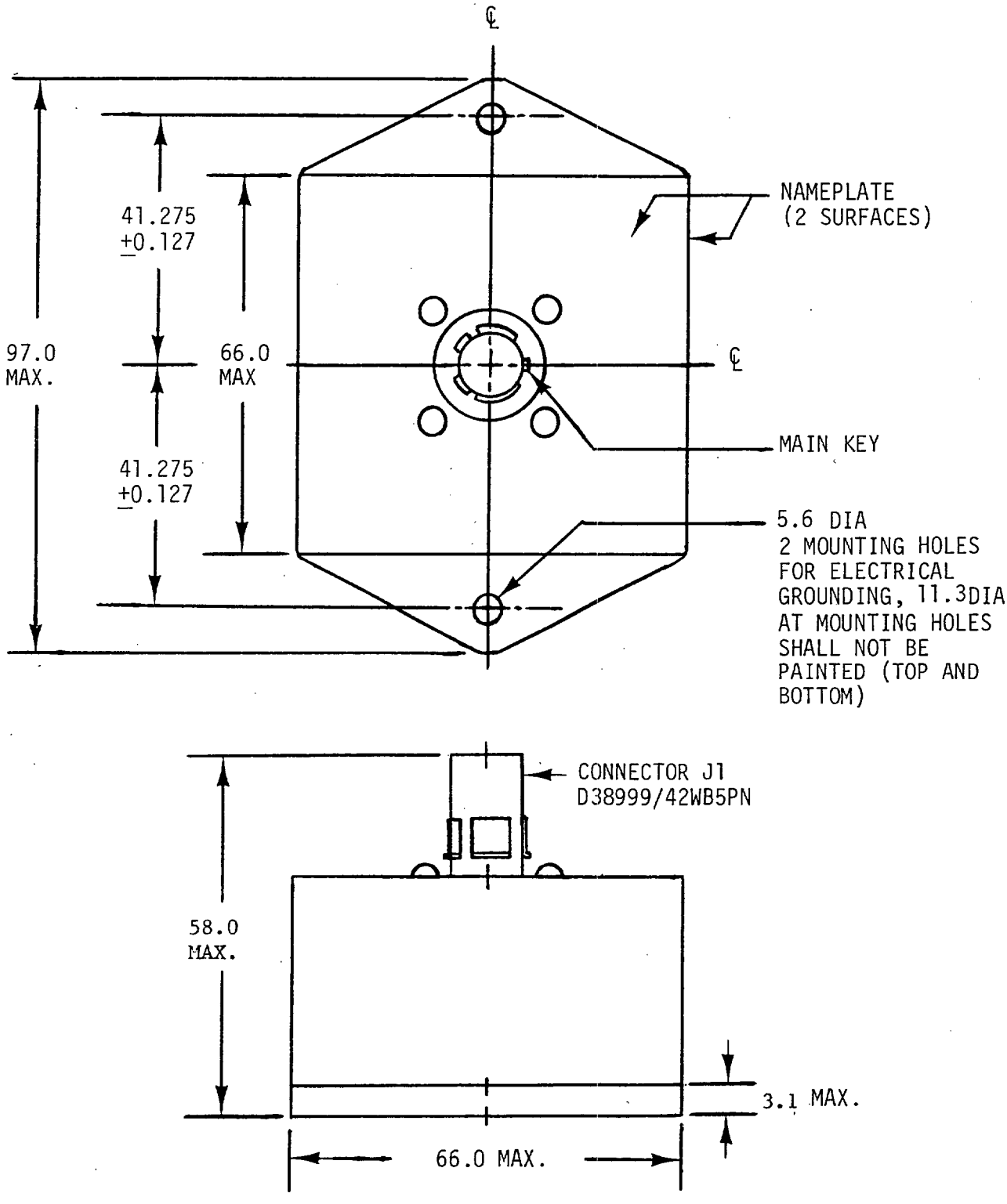
Relay Control Unit, Battery, Aircraft

Aircraft Battery Relay

Electrical Bus

Preparing Activity
Navy - AS
(Project No. 6140-N685)

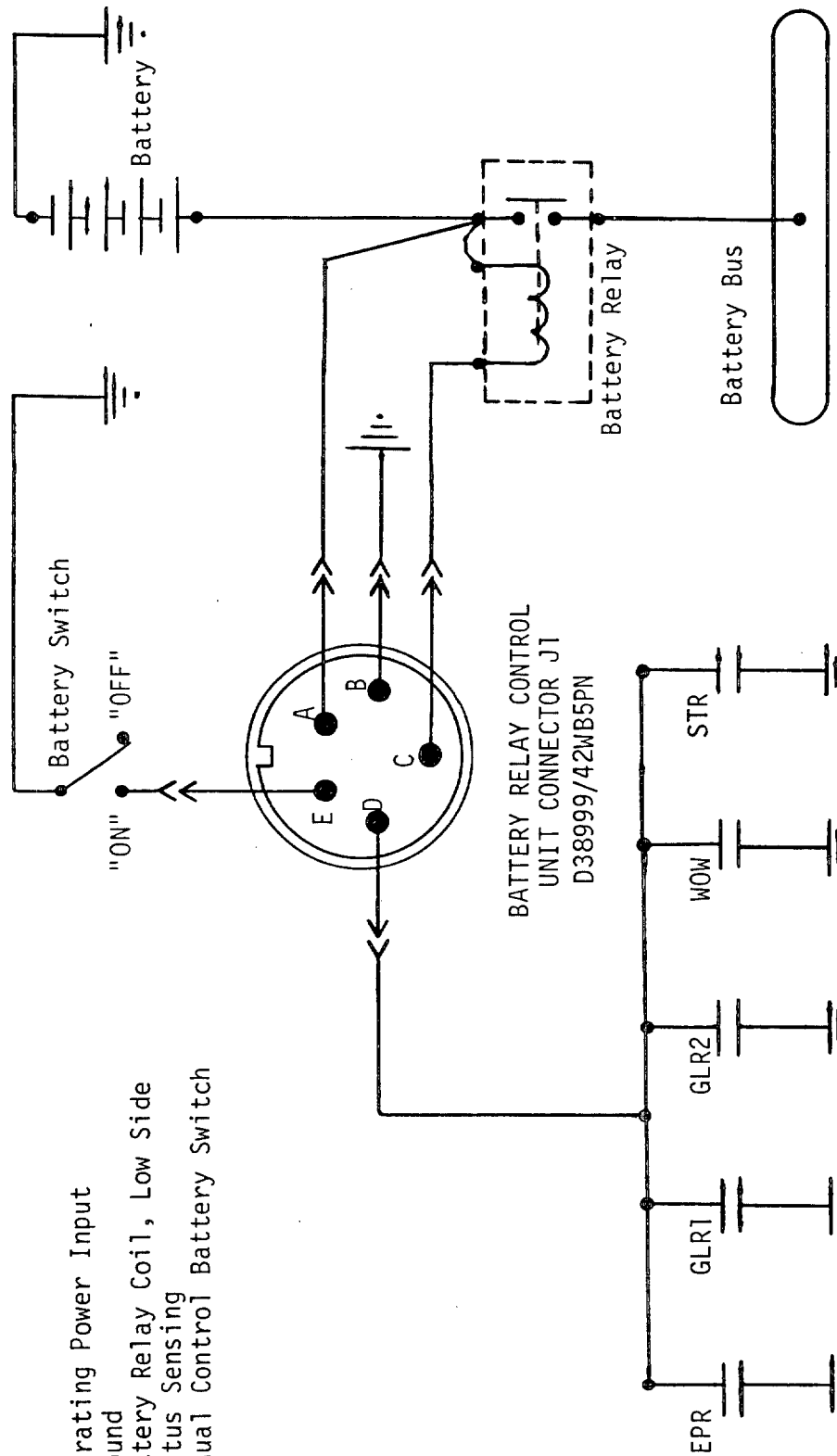
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ALL DIMENSIONS ARE IN MILLIMETERS.
TOLERANCE: ±0.4 UNLESS OTHERWISE SHOWN.
PART NUMBER D85584-1

FIGURE 1. BATTERY RELAY CONTROL UNIT DIMENSIONS

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NOTE:

J1-A Operating Power Input
J1-B Ground
J1-C Battery Relay Coil, Low Side
J1-D Status Sensing
J1-E Manual Control Battery Switch

NOTE:

EPR - External power relay is closed when the aircraft is receiving external electrical power.
GLR - Generator line relays are closed when aircraft generators are providing power.
WOW - Weight on wheels relay is closed when there is no weight on landing gear wheels.
STR - Start relay closed during engine starting.

FIGURE 2. DIAGRAM OF AIRCRAFT WIRING CONNECTIONS TO BATTERY RELAY CONTROL UNIT CONNECTOR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-85584A	BATTERY RELAY CONTROL UNIT, AIRCRAFT

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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